

FHK-Series

Heavy-duty Metal Sheath Element Electric Portable Infrared Heaters



FHK-2 series

FHK-4 series

FHK-13 series

FHK-6 series

WARNING

IMPORTANT SAFETY INFORMATION INSIDE

- Serious injury or death possible
- Read, understand, and follow all safety information and instructions in this manual before using or servicing this product.
- Retain these instructions for future reference.



If you have questions about the product you have purchased or would like to leave us feedback please contact us via our website www.tpicorp.com or by calling 1-800-251-0382.

REFER TO COMPLETE
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INFORMATION ON
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ATTENTION:

The table to the right provides definitions of the signal words that can be found throughout this manual. These signal words are used to express the severity of the hazard at hand. The signal words are generally used in conjunction with safety symbols that correspond to the text for that particular hazard. As you read this manual, refer back to this table when you are unsure of the signal word definition.

SIGNAL WORD DEFINITIONS	
	DANGER indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
	WARNING indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
	CAUTION indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.
	CAUTION used without the safety alert symbol indicates a potentially hazardous situation which, if not avoided, may result in property damage.
As defined in ANSI Z535.4-2002	

THANK YOU!!

Thank you very much for selecting Fostoria's portable electric infrared heating equipment for your comfort heating needs. These products were engineered with the most reliable components and materials available and are equipped with features to assure ease of installation and maintenance.

The warmth you will now enjoy when using this heating equipment is created by a heat source; a metal-sheathed element; that emits infrared energy in the form of heat, like the sun. This is an economical way to heat because it heats people and objects, not the air, so you don't need to heat a large area to feel warm.

CHECKING YOUR SHIPMENT

- ❑ Upon receipt of your shipment, check all cartons for visible damage.
- ❑ Claims for damaged material or shortages that were not evident upon receipt of shipment must be reported to carrier and TPI Corporation's customer service (800-251-0382) immediately.
- ❑ Any accessory items ordered for the heater will be shipped in separate cartons.

GENERAL DESCRIPTION & USE

WARNING

EXPLOSION HAZARD FIRE HAZARD BURN HAZARD

- **Serious injury or death may occur.**
- **Not for indoor residential use.**
- **Do not use outdoors in wet conditions and in unprotected locations.**
- **Use for comfort heating only.**
- **Do not use in locations containing hazardous or explosive atmospheres.**
- **Comply with "REQUIRED CLEARANCES" table at right for minimum clearances to combustible materials.**

The Fostoria FHK-series is a line of heavy-duty portable heaters designed to provide electric infrared heat when permanently installed heaters are not practical or economical. They are easily transported to the work site and connected to disconnect switches or lighting centers. The units are suited to a wide variety of indoor and protected outdoor commercial or industrial uses. They operate cleanly, emitting no fumes or odors.

Applications where combustible materials and/or personnel will be located near these heaters require the addition of a ground-fault sensing switch as part of the electrical circuit. See the INSTALLATION PLANNING section of this manual for more details. Any installation of these heaters must also adhere to the minimum clearance information below.

MODEL	REQUIRED CLEARANCES TO COMBUSTIBLES FROM FRONT OF HEATER
FHK-212-1CA	2 Feet
FHK-2	3 feet
FHK-4	3 Feet
FHK-6	4 Feet
FHK-13	7 Feet

SPECIFICATIONS

FHK-2 & FHK-4 Series:

- ❑ Powder-coated steel base and large carry handle
- ❑ Gold-anodized aluminum housing for good reflectivity and corrosion resistance
- ❑ Housing is of double-wall construction for reduced surface temperatures
- ❑ Features aluminum extrusion and element support clip for high-vibration areas and rugged environments
- ❑ U-shaped metal-sheath elements are factory-installed and mounted on a cast aluminum terminal box
- ❑ 120V model (FHK-212-1CA) is equipped with 15ft cordset with NEMA 5-15 molded plug

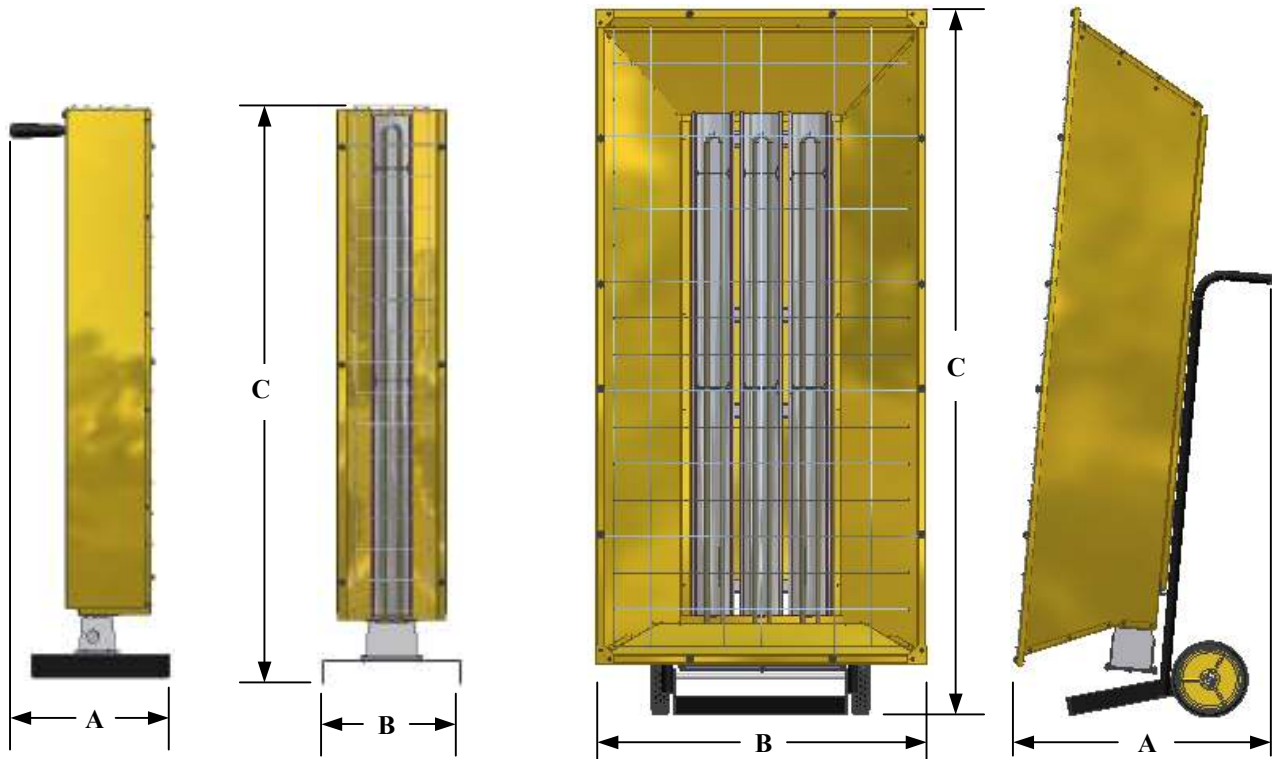
FHK-6 & FHK-13 Series:

- ❑ Heaters are mounted on steel handcarts with 6-inch wheels, which allow easy transport to the job site.
- ❑ Gold-anodized aluminum housing for good reflectivity and corrosion resistance
- ❑ Housing is of double-wall construction for reduced surface temperatures
- ❑ Features aluminum extrusion and element support clip for high-vibration areas and rugged environments
- ❑ U-shaped metal-sheath elements are factory-installed and mounted on a corrosion protected steel terminal box
- ❑ Model FHK-624-1CA is equipped with a 15ft, type S, 3-wire cable with a #10-3 dryer plug

Heater Models	Voltage And Phase	Amps 1ph.	Amps 3ph.	Number Of Elements	Watts	Btu's/Hr.	Dimensions (in.)		
							A	B	C
FHK-212-1CA	120; 1ph	15.0	n/a	1	1800	6,142	13-1/2	11-1/2	28-1/2
FHK-220-1A	208; 1ph	9.7	n/a		2000	6,824			
FHK-224-1A	240; 1ph	8.4	n/a						
FHK-227-1A	277; 1ph	7.3	n/a						
FHK-248-1A	480; 1ph	4.2	n/a						
FHK-257-1A	600; 1ph	3.5	n/a						
FHK-420-1A	208; 1ph	21.7	n/a	1	4500	15,355	13-1/2	11-1/2	48-3/4
FHK-424-1A	240; 1ph	18.8	n/a						
FHK-427-1A	277; 1ph	16.3	n/a						
FHK-448-1A	480; 1ph	9.4	n/a						
FHK-457-1A	600; 1ph	7.8	n/a						
FHK-620-3A	208; 1 or 3ph	28.9	16.6	3	6000	20,473	17	19-3/4	37-1/2
FHK-624-3A	240; 1 or 3ph	25.0	14.4						
FHK-624-1CA	240; 1ph	25.0	n/a						
FHK-627-3A	277; 1ph	21.7	n/a						
FHK-648-3A	480; 1 or 3ph	12.5	7.2						
FHK-657-3A	600; 1 or 3ph	10.4	6.1						
FHK-1320-3A	208; 3ph	n/a	37.4	3	13,500	46,064	19	27-1/2	59
FHK-1324-3A	240; 3ph	n/a	32.4						
FHK-1327-3A	277; 1ph	48.8	n/a						
FHK-1348-3A	480; 3ph	n/a	16.2						
FHK-1357-3A	600; 3ph	n/a	13.6						

Please see illustrations on next page regarding dimensions in table.

SPECIFICATIONS (cont.)



FHK-2 & 4 series

FHK-6 & 13 series

INSTALLATION PLANNING

Fostoria's metal-sheath infrared heaters are equipped with heating elements that are manufactured using materials with exceptional mechanical and thermal performance and are built with precise process control to provide rugged and reliable infrared performance in challenging environments. It is our understanding that these precision characteristics can change with time, specifically in environments with high humidity or where mechanical damage can occur, causing the internal electrical insulation to weaken. Use of Fostoria's GF-SK-1 ground fault sensing switch will monitor this gradual change and interrupt power to the heaters prior to any arc-fault type failures, thereby protecting against a possible fire or burn hazard. **Any installation where this is a concern must be protected by using the GF-SK-1 switch.**



WARNING

FIRE HAZARD BURN HAZARD

- To prevent arcing type faults, premature element failure and potential fire or burn hazard, use Fostoria model GF-SK-1 ground fault switch with these heaters.
- Read and follow all instructions provided for proper installation of heaters and GF-SK-1.

Options for using the GF-SK-1 with Fostoria FHK-Series heaters:

Mounting Option	Heater Compatibility
Mounting GF-SK-1 directly onto heater	FHK-6xx, FHK-13xx (cannot be mounted to FHK-2xx and FHK-4xx)
Wall-mounting the GF-SK-1 (see explanation below)	FHK-2xx, FHK-4xx, FHK-6xx, FHK-13xx

The GF-SK-1 can monitor a single heater as denoted above but the most cost effective arrangement of the system is realized when monitoring a circuit of multiple heaters. The total number of heaters that can be monitored by one GF-SK-1 is dependant upon the heater models selected and the supply voltage. Using the total watts of the selected heaters to be monitored, the known supply voltage (be certain the heater voltage rating matches the supply voltage), and the phase of the supply, the total load or “amps” can be calculated. The GF-SK-1 is capable of monitoring a heater circuit of up to 50 amps maximum. **A licensed electrician or design engineer must do the necessary calculations.**

WARNING

FIRE HAZARD

- Serious injury or death may occur.**
- Heater circuit load calculations and circuit design must be done by a licensed electrician or engineer.**

Preparations for Wiring Heaters

CAUTION

Over-voltage is a major cause of element failure with these heaters. Make certain the power supply voltage is no more than 2% above the rated voltage of the heater being utilized.

Installer Qualifications

The installation and wiring of Fostoria FHK heaters must be performed by a licensed electrician.

Installer Responsibility

The National Electric Code (NEC) and local codes and ordinances together with specifications provided by Fostoria comprise the information needed for proper installation. The installer must furnish all materials that have not been purchased from Fostoria or its representatives. It is the installer’s responsibility that the materials and methods of installation result in a job that is workmanlike and compliant with all applicable codes.

- ❑ The maximum ampacity rating for an infrared heating circuit is 50A.
- ❑ The opening in the terminal box for branch circuit supply wiring on FHK-2 models is sized for 1/2-inch trade size cable connectors. Openings in terminal boxes for supply wiring on FHK-4, FHK-6 and FHK-13 models are sized for 3/4-inch trade size cable connectors.
- ❑ Cable connectors and cables are not provided with these heaters. (See Specifications on page 4 for exceptions.)

INSTALLATION PLANNING (cont.)

The following tables are provided to assist the licensed electrician in sizing the electric cables and/or plugs needed for each FHK heater model. Since only models FHK-212-1CA and FHK-624-1CA are equipped with cables and plugs, these items for all other models will have to be obtained through a local supplier or purchased separately from Fostoria. (See Table B.)

All supply cables must be copper, rated 90°C and comply with all local codes and the National Electric Code (NEC).

All recommended wire sizes assume a 50ft. maximum wire length.

Table A

Heater Model	KW	VOLTS	AMPS		Type SO Cable Size*		Plug Size 3-PH	Plug Size 1-PH
			3-PH	1-PH	3-PH	1-PH		
FHK-220-1A	2	208	N/A	9.7	N/A	16/3	N/A	NEMA L6-15
FHK-224-1A	2	240	N/A	8.4	N/A	16/3	N/A	NEMA L6-16
FHK-227-1A	2	277	N/A	7.3	N/A	16/3	N/A	NEMA L7-17
FHK-248-1A	2	480	N/A	4.2	N/A	16/3	N/A	NEMA L8-20
FHK-257-1A	2	600	N/A	3.5	N/A	16/3	N/A	NEMA L9-20
FHK-420-1A	4.5	208	N/A	21.7	N/A	10/3	N/A	30A/250V/3 Wire
FHK-424-1A	4.5	240	N/A	18.8	N/A	12/3	N/A	20A/250V/3 Wire
FHK-427-1A	4.5	277	N/A	16.3	N/A	12/3	N/A	20A/480V/3 Wire
FHK-448-1A	4.5	480	N/A	9.4	N/A	14/3	N/A	10A/480V/3 Wire
FHK-457-1A	4.5	600	N/A	7.8	N/A	14/3	N/A	10A/600V/3 Wire
FHK-620-3A	6	208	16.6	28.9	12/4	8/3	20A/250V/4 Wire	30A/250V/3 Wire
FHK-624-3A	6	240	14.4	25.0	14/4	10/3	20A/250V/4 Wire	30A/250V/3 Wire
FHK-627-3A	6	277	N/A	21.7	N/A	10/3	N/A	30A/480V/3 Wire
FHK-648-3A	6	480	7.2	12.5	14/4	14/3	20A/480V/4 Wire	20A/480V/3 Wire
FHK-657-3A	6	600	6.1	10.4	14/4	14/3	20A/600V/4 Wire	20A/600v/3 Wire
FHK-1320-3A	13.5	208	37.4	N/A	6/4	N/A	50A/250V/4 Wire	N/A
FHK-1324-3A	13.5	240	32.4	N/A	8/4	N/A	50A/250V/4 Wire	N/A
FHK-1327-3A	13.5	277	N/A	48.8	N/A	4/3	N/A	50A/480V/3 Wire
FHK-1348-3A	13.5	480	16.2	N/A	12/4	N/A	20A/480V/4 Wire	N/A
FHK-1357-3A	13.5	600	13.6	N/A	12/4	N/A	20A/600V/4 Wire	N/A

Table B

ELECTRIC CABLES			
SIZE/TYPE	PART NO.	SIZE/TYPE	PART NO.
16/3 SO	08804300	8/3 SO	08805200
14/3 SO	08804600	8/4 SO	08805100
14/4 SO	08804900	6/4 SO	03164001
12/3 SO	08804500	4/3 SO	08805300
12/4 SO	03164201	2/3 SO	08804700
10/3 SO	08804400		

For All Heaters



⚠ WARNING

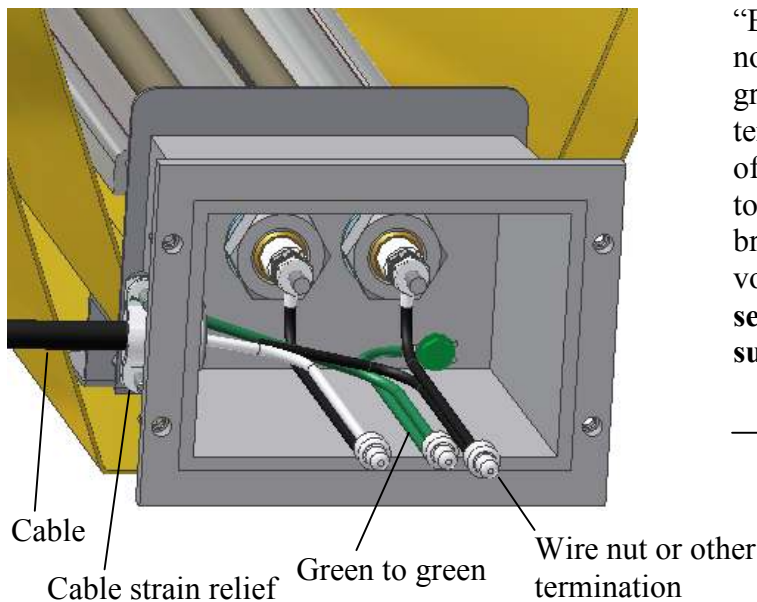
ELECTRICAL SHOCK HAZARD

- **Serious injury or death may occur.**
- **Disconnect heater and lock out from electrical supply before installing or servicing.**
- **Always securely connect green wire on heater to bonding (green) wire from electrical supply.**

The FHK-series heaters do not have any onboard controls and therefore all permanently connected heaters must be used in conjunction with the applicable Fostoria FPC-series control center or be wired to a listed, properly rated safety switch for on/off control. Always disconnect the power at the circuit breaker or disconnect switch before installing electric power to the heater. If the breaker or disconnect switch cannot be seen from where you will be working, lock it in the open position and tag it to prevent unexpected application of power. **Failure to do so could result in fatal electric shock.** Do not depend on a thermostat or other switch as the sole means of turning off power to the heater.

To disconnect power to heaters using wall receptacles and plugs, simply unplug the heater.

“Bonding” is the intentional electrical connection of all non-current carrying metal parts to the equipment grounding (green) conductor of the heater and then terminating this green conductor to the green conductor of the power supply to form a low-impedance path back to the power supply so that safety devices (circuit breakers, fuses) can quickly remove dangerous touch voltage from these parts when a fault occurs. **Always securely connect the green conductor from the power supply to the green conductor of the heater.**



Wiring of FHK-2 and FHK-4 Series Heaters

Since all FHK-2 and FHK-4 heaters (except FHK-212-1CA) are received without an electric cable, the following steps must be taken before attaching the heater to the base:

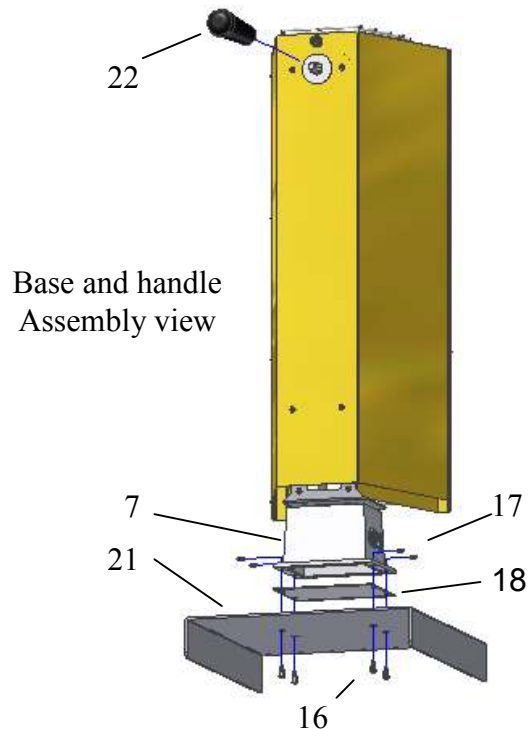
1. Install a standard or liquid-tight cable connector into the hole in the side of the terminal box. (See page 6 for appropriate cable connector sizes.)
2. Place the properly sized supply cable through the fitting. See page 7 for cable and plug size specifications or refer to the National Electric Code (NEC).
3. Connect supply cable leads to the element lead wires in the terminal box with wire nuts or other safe mechanical means. Black and white cable leads can be connected to either element lead wire. Connect green lead from supply cable to green (bonding) lead in the terminal box. (See illustration shown above.)

Continued on next page

INSTALLATION – WIRING (cont.)

4. Position the rubber gasket (item 18) and the base (item 21) on the bottom of the terminal box (item 7), aligning the slotted holes. Fasten the base and terminal box together with the screws and nuts (items 16 & 17) provided. (See page 14 (FHK-2) or 15 (FHK-4) for item number list.)
5. Screw handle (item 22) onto the 3/8" bolt on the back of the housing.
6. Connect heater to a suitable electric circuit.

Note: Handle, gasket, bolts, and nuts are in hardware package 3917900.



INSTALLATION – WIRING (cont.)

Wiring of FHK-6 and FHK-13 Series Heaters:

All of these three-element models are factory-wired for three-phase operation. Voltages other than 277V are wired as shown in illustration 1. The 277V models are wired as shown in illustration 2. To connect the appropriate electrical service to your heater, follow the steps below.

1. Remove the cover and gasket from the bottom of the terminal box, save screws.
2. Select the properly sized cable from tables on page 7 and route it through the cable connector and into terminal box. Use a 3/4 inch cable connector* to secure the cable to the terminal box.
3. Use standard clamp type connectors, or if the heater is to be used in damp locations, use liquid tight cable connectors*.
4. Connect cable leads to the 6-inch element lead wires inside the terminal box, with wire nuts or other safe mechanical means.
5. Reattach the cover and gasket removed in step 1.
6. If a plug* is to be used, blade configuration must be selected per local codes and the NEC. Reference plug sizes on page 7, Table A.
7. Connect the heater to a suitable electrical circuit.

* These items are not supplied with the heaters and must be purchased locally.

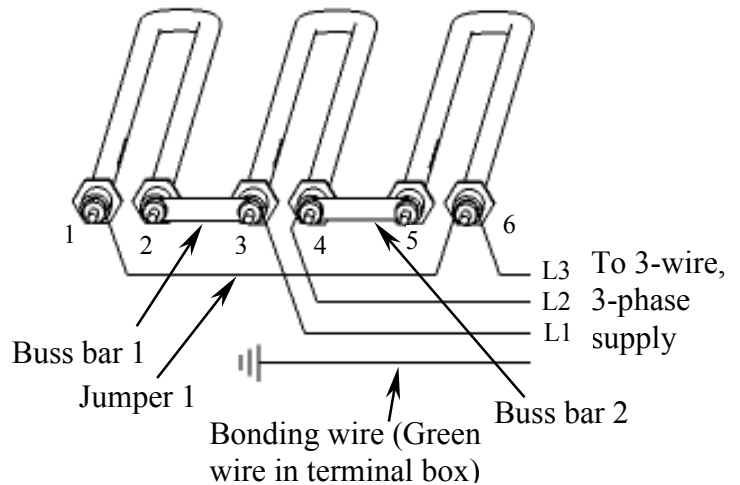


Illustration 1

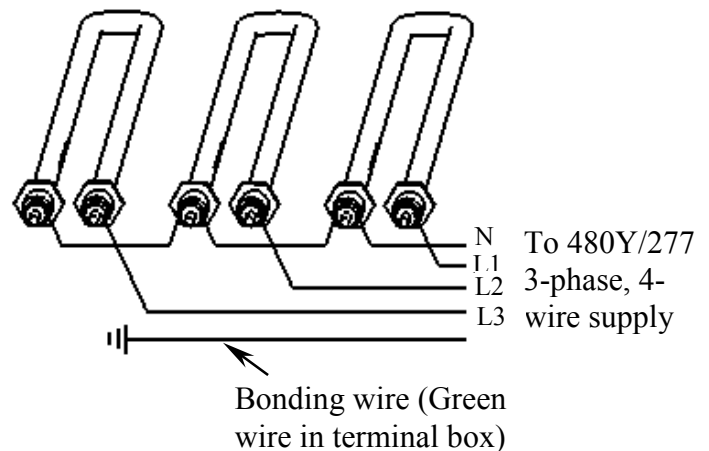


Illustration 2

3-Phase to Single-Phase Conversion:

This conversion is allowable on FHK-6 series models only. To make this conversion, the licensed electrician must take the arrangement in illustration 1 and revise it to become as shown in illustration 3 by following the steps below.

1. Remove pigtail L3 from post 6 (illustration 1).
2. Move one end of Jumper 1 from post 6 to post 5.
3. Install Jumper 2** on posts 2 and 6.
4. Do not move Buss bars 1 and 2.
5. **Replace supply wires with heavier gauge to meet additional amp load.** Reference table on page 7.

** Jumper 2 (customer supplied) must be 12 AWG, 200° C rated type SEW-2 or equivalent, UL style 3074 wire, at least 10 inches long. This jumper (p/n 3160609) can be purchased from the factory.

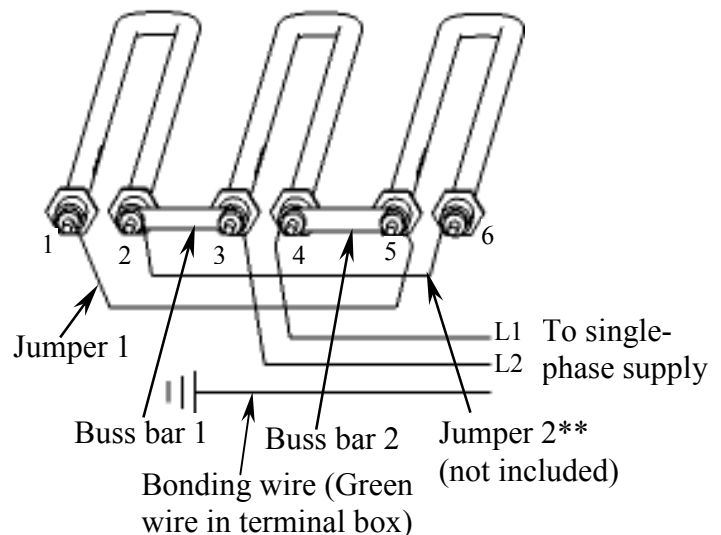


Illustration 3

⚠ WARNING

**FIRE HAZARD
BURN HAZARD**



- **Serious injury or death may occur.**
- **Always operate heaters on level surfaces.**
- **Allow heater to fully cool after power is removed before touching or working on it.**
- **Do not leave heaters unattended while in operation**
- **Maintain minimum clearances to combustibles as shown in table below.**

When power is provided to these heaters the heater elements will take several minutes to achieve full infrared output that can be confirmed by an orange to dull red glow on the element sheath. The element sheath under this condition is operating between 1,500 and 1,600° F emitting medium wavelength (approx. 3 microns or millionths of a meter) infrared energy. This energy, that can also be considered “radiation” is not harmful, yet is very effective at heating objects and surfaces that are within the line-of-sight of the elements and the heater’s reflective surfaces.

When power is provided or removed from these heaters, you may hear unfamiliar sounds that are caused by the expansion and contraction of metal components in the heater. This is normal.

There are no on/off switches, tip-over switches or over-temperature safety cut out devices included on any of these heaters. See the Accessories section of this manual for more details.

MODEL	REQUIRED CLEARANCES TO COMBUSTIBLES FROM FRONT OF HEATER
FHK-212-1CA	2 Feet
FHK-2	3 feet
FHK-4	3 Feet
FHK-6	4 Feet
FHK-13	7 Feet

CAUTION

If any metal-sheath element has an obvious “bowing” or sagging condition during operation, an over-voltage / over-temperature condition may exist. De-energize the heater immediately and allow to fully cool. Replace all elements that have been distorted in any way. Before returning power to heaters, make certain the power supply voltage is no more than 2% above the rating on the heater data label.

OPERATION (cont.)

Ground-Fault Monitoring and Operation:

As explained in the “installation planning” section of this manual, most, if not all, applications using metal-sheath heaters will require the use of the GF-SK-1 ground-fault sensor switch. Refer to the instructions provided with this switch for proper installation.

If, during heater operation, the GF-SK-1 “trips” and interrupts power to your heater or heater circuit, then one or more of the metal-sheath elements in the heater or heater circuit being monitored has a reduced insulation resistance value that may be due to an element that is about to fail. Have a licensed electrician follow the warnings and instructions in the “operation” section of the GF-SK-1 manual.

	WARNING
	FIRE HAZARD BURN HAZARD • To prevent arcing type faults, premature element failure and potential fire or burn hazard, use Fostoria model GF-SK-1 ground fault switch with these heaters. • Read and follow all instructions provided for proper installation of heaters and GF-SK-1.

MAINTENANCE

Pre-Season Maintenance and Annual Inspection

To ensure your safety and years of trouble-free operation from the heaters, periodic service and inspections must be done by a trained maintenance person or licensed electrician.

To obtain maximum performance from your heater(s) each year, we recommend the following be performed at the start of the heating season:

1. Clean housing reflector surfaces with a damp cloth.
2. Blow or dust off the elements and inner reflectors.
3. Repair or replace damaged power cables.
4. DO NOT hose-down these heaters.

Storage

When storing the heaters for long periods they should be covered to avoid accumulation of dust and other contaminants. Keep them in dry areas away from dripping or standing water and heavy humidity.

CAUTION

- Do not store in a damp location.

ELEMENT & REFLECTOR REPLACEMENT: FHK-2 & FHK-4



⚠ WARNING

ELECTRICAL SHOCK HAZARD

- Serious injury or death may occur.
- Disconnect heaters from electrical supply before replacing parts.



CAUTION

Use only Fostoria replacement elements of the same wattage, voltage and part number as the original element. Reference tables on replacement parts pages later in this manual. Always install new reflector included with each replacement element.

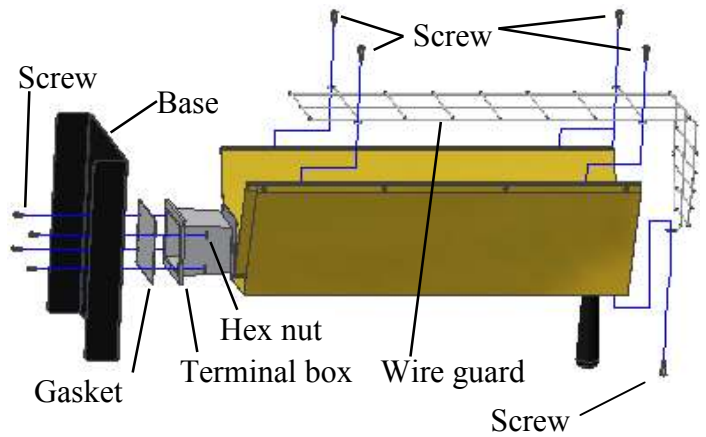


Illustration 4

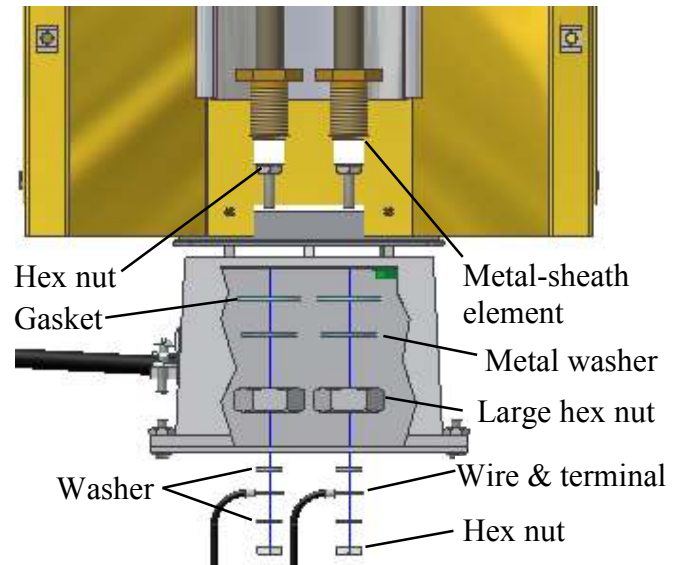


Illustration 5

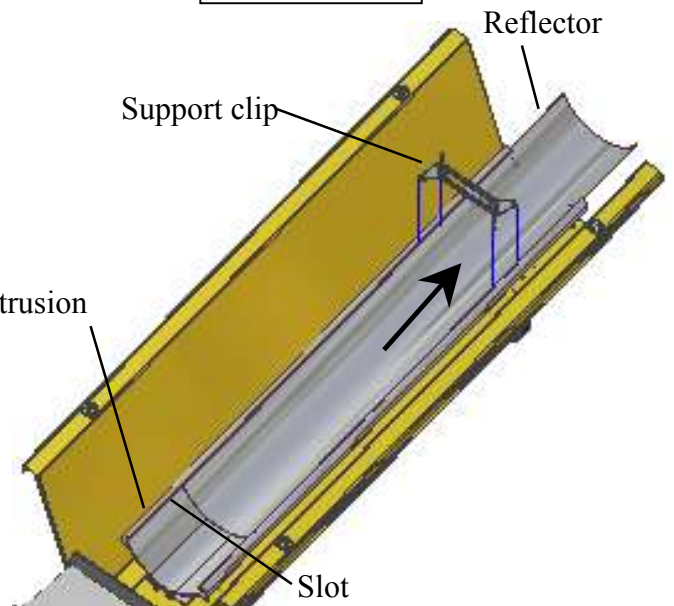


Illustration 6

1. DISCONNECT POWER FROM THE ELECTRICAL SUPPLY CABLE OF THE PORTABLE HEATER BEING REPAIRED.
2. Position heater on workbench with reflector opening pointing UP.
3. Remove wire guard by taking out the (5) screws shown in Illustration 4 with a 5/16" nut driver, save hardware.
4. Remove element support clip (2 on FHK-4 models), save clip(s).
5. Remove base and gasket from terminal box on heater; save base, gasket and hardware.
6. Disconnect the 2 wires from heating element by loosening the 3/8" hex nuts at the bottom of the rod.
7. Remove old element by removing the large hex nuts, metal washers and gaskets, discard all.
8. Slide old reflector out of the extrusion and discard.
9. Peel protective film from new reflector, and then place it in the extrusion. Make sure reflector is secured in slots in extrusion.
10. Install new element in terminal box, using hardware included. See Illustration 5.
11. Reinstall element support clip(s), removed in Step 4, into extrusion.
12. Connect the two wires to the element as shown in Illustration 5. Make sure that bare metal wire terminals from different posts do not come in contact with each other.
13. Reattach base and gasket to terminal box with hardware removed in Step 5.
14. Reattach wire guard to housing with screws removed in Step 3.
15. Heater is now ready for normal operation.

ELEMENT REPLACEMENT: FHK-6 & FHK-13



⚠ WARNING

ELECTRICAL SHOCK HAZARD

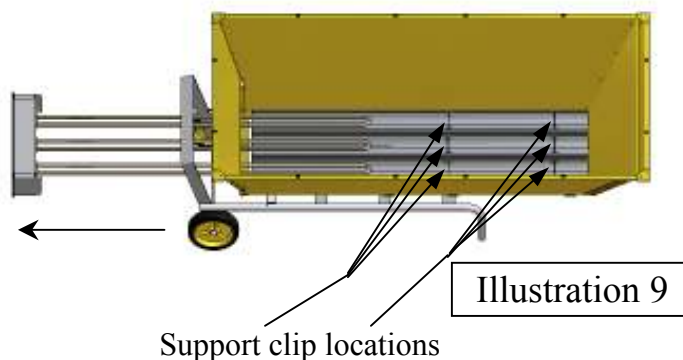
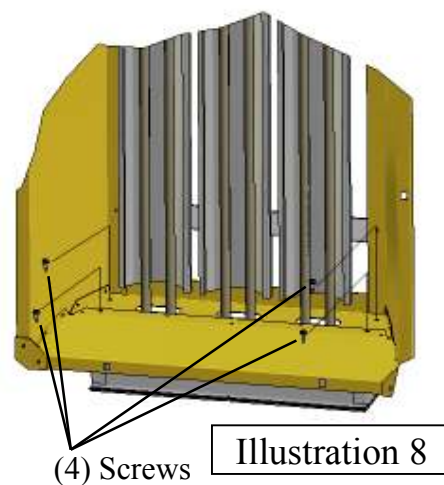
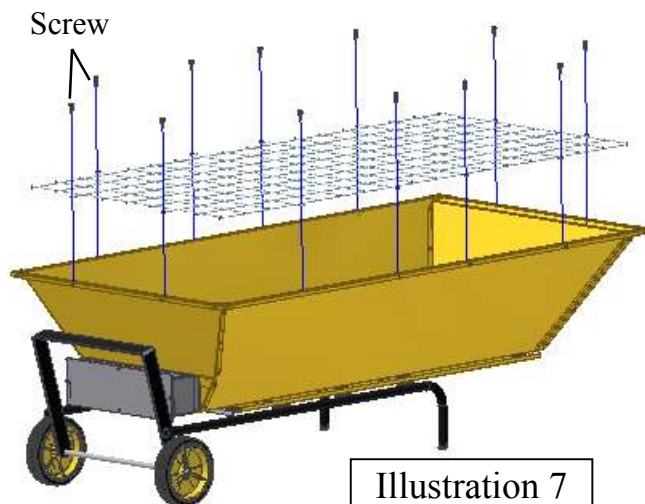
- **Serious injury or death may occur.**
- **Disconnect heaters from electrical supply before replacing parts.**

CAUTION

Use only Fostoria replacement elements of the same wattage, voltage and part number as the original element. Reference tables on replacement parts pages later in this manual.

Always install new reflector included with each replacement element.

1. DISCONNECT POWER FROM THE ELECTRICAL SUPPLY CABLE OF THE PORTABLE HEATER BEING REPAIRED.
2. Position heater with reflector opening pointing UP.
3. Remove the wire guard from the heater to access the fasteners that attach the terminal box. Remove and save the (12) screws shown in Illustration 7 with a 5/16" nut driver.
4. Remove and save the (4) screws that attach the terminal box to the heater, with a 1/4" nut driver. See Illustration 8.
5. Grasp terminal box assembly and firmly pull in the direction shown in Illustration 9. The elements have been fed through some support clips inside the heater housing. These clips will cause some drag or resistance to removal. There will be some unpleasant noise as the elements slide through the clips.
6. Remove cover and gasket from terminal box; save them and all screws.
7. Disconnect the wires from only the heating element to be replaced by loosening the 3/8" hex nuts at the bottom of the element similar to Illustration 5.
8. Remove old element by removing the large hex nuts, metal washers and gaskets, discard all.
9. Install new element in terminal box, using hardware included. See Illustration 5.
10. Reconnect the wires to the element as they were before removal. Make sure that bare metal wire terminals from different posts do not come in contact with each other.
11. Reattach base, gasket and screws to terminal box.
12. Before reinstalling the heating elements, the old reflector should be replaced with the new one received with the new heating element. See next page for Reflector Replacement.
13. After completing all steps in Reflector Replacement, place the heating element assembly back into the housing and secure with screws removed in Step 4.
14. Reattach wire guard with screws removed in Step 3.
15. Heater is now ready for normal operation.



REFLECTOR REPLACEMENT: FHK-6 & FHK-13



WARNING

ELECTRICAL SHOCK HAZARD

- **Serious injury or death may occur.**
- **Disconnect heaters from electrical supply before replacing parts.**

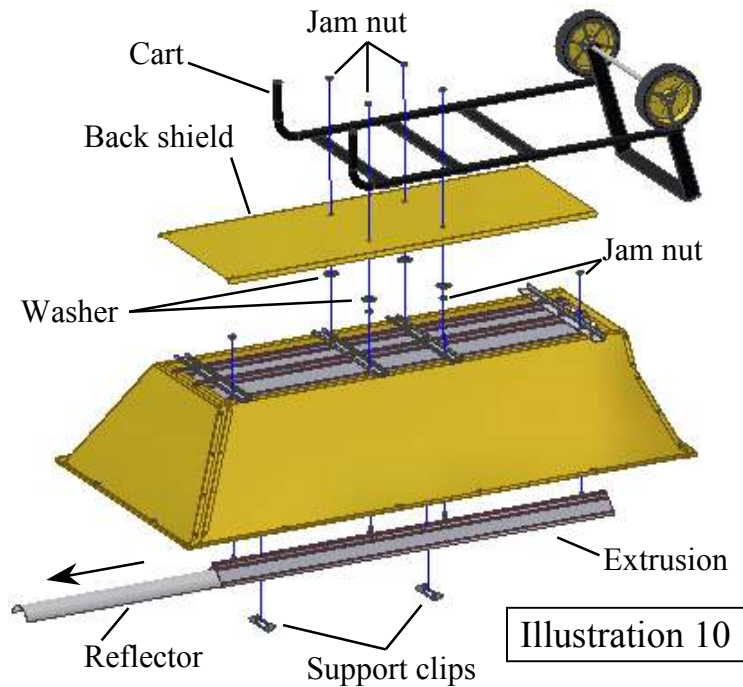


Illustration 10

To replace the reflector:

1. Lay heater face down and remove the cart from the heater by taking off the (4) jam nuts as shown in Illustration 10.
2. Lift the back shield and (4) washers off the heater.
3. Remove the (2) support clips from the front of the reflector.
4. Remove (4) jam nuts that secure the extrusion to the housing. The extrusion and reflector will now come out of the housing.
5. Slide the reflector out of the extrusion and discard.
6. Peel protective film from new reflector, and then place the reflector in the extrusion. Make sure it is secured in slots in extrusion.
7. Reattach extrusion, back shield and cart, using hardware removed in previous steps.
8. Replace support clips in the front of the reflector.
9. Finish reassembly of heater by completing Steps 13 and 14 on page 14.

ACCESSORIES

The kits shown below are optional controls for the FHK Series heaters.



DC-SK-1
Disconnect switch
Part No.: 05893702

- Wall mountable for all FHK models
- Mounts directly on FHK-6 & FHK-13 models only



TO-SK-1
Tip-over switch
Part No.: 04858102

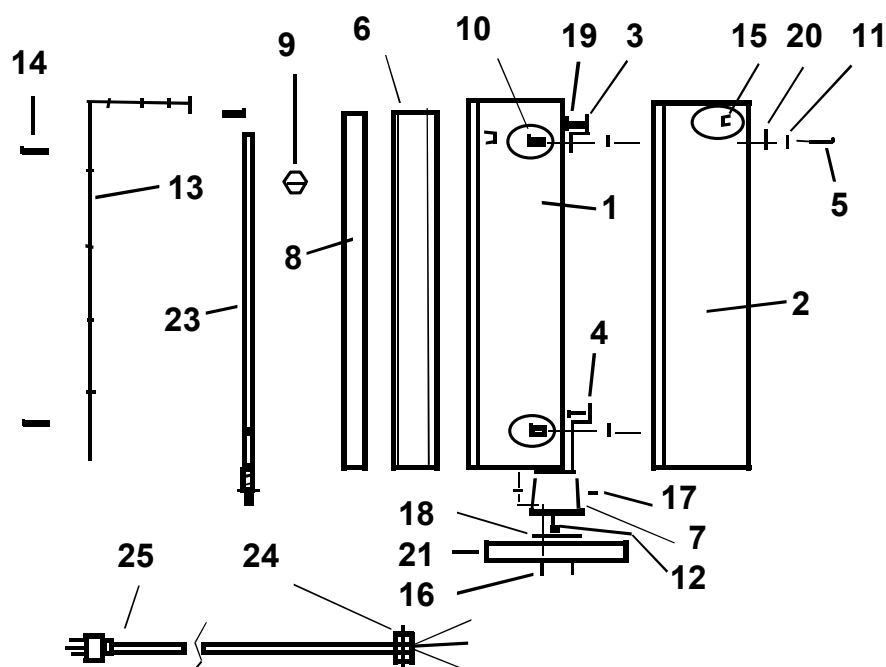
- Mounts directly on FHK-6 & FHK-13 models only



GF-SK-1
Ground fault sensor switch
Part No.: 04860202

- Wall mountable for all FHK models
- Mounts directly on FHK-6 & FHK-13 models only

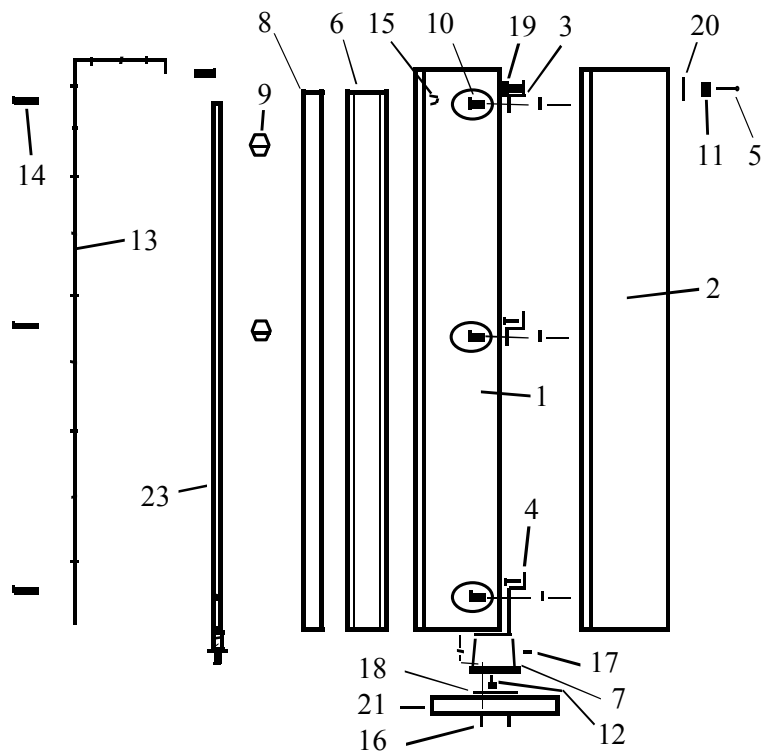
FHK-2 EXPLODED VIEW AND REPLACEMENT PARTS LIST



ITEM NO.	DESCRIPTION	PART NUMBER	QTY.
1	Side baffle	8022707	2
2	Back shield	8022607	1
3	Mounting bracket	8023000	1
4	Mounting brace	8023100	1
5	Pop rivet	3482200	18
6	Extrusion	8041700	1
7	Terminal box	8040501	1
8	Reflector	8048900	1
9	Element support	8024100	1
10	Screw 3/8-16 x 1/2	3474300	2
11	Nut 3/8-16	3429100	3
12	Screw 10-32 x 3/8	3475000	1
13	Wire guard	8022329	1
14	Screw 10-32 x 3/4	3424956	5
15	Nut self-retaining 10-32	3403900	5
16	Screw 8-32 x 1/2	3427456	4
17	Nut 8-32	3405156	4
18	Gasket	8040700	1
19	Screw 3/8-16 x 1	3493900	1
20	Washer fender 3/8	3478600	1
21	Base	8042921	1
22	Handle	8041800	1
23	Metal-sheath element	See Table 1	1
24	Cable connector 1/2" *	7714114	1
25	14-3 Cordset *	3175600	1

Table 1		
MODEL	VOLTS	METAL-SHEATH ELEMENT PART NUMBER
FHK-212-1CA	120	04458702
FHK-220-1A	208	04458302
FHK-224-1A	240	04457902
FHK-227-1A	277	04458402
FHK-248-1A	480	04458002
FHK-257-1A	600	04458802

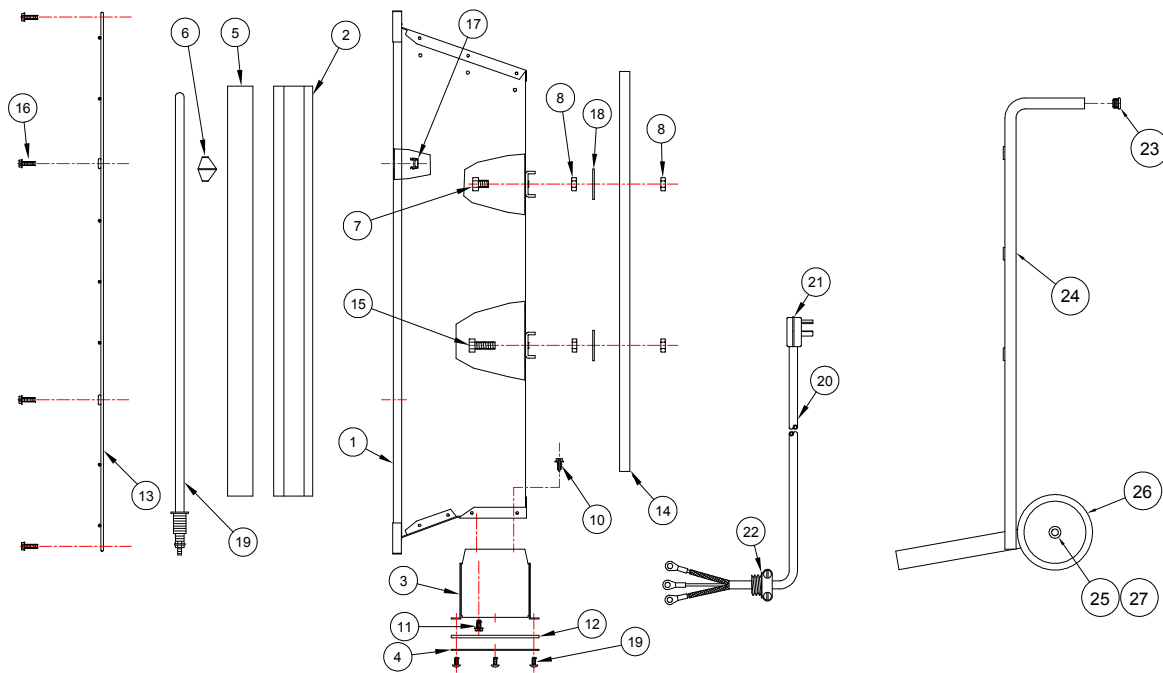
FHK-4 EXPLODED VIEW AND REPLACEMENT PARTS LIST



ITEM NO.	DESCRIPTION	PART NUMBER	QTY.
1	Side baffle	8022907	2
2	Back shield	8022807	1
3	Mounting bracket	8023000	2
4	Mounting brace	8023100	1
5	Pop rivet	3482200	26
6	Extrusion	8042400	1
7	Terminal box	8040502	1
8	Reflector	8049000	1
9	Element support	8024100	2
10	Screw 3/8-16 x 1/2	3474300	3
11	Jam nut 3/8-16	3429100	4
12	Screw 10-32 x 3/8	3475000	1
13	Wireguard	8022429	1
14	Screw 10-32 x 3/4	3424956	7
15	Nut self-retaining 10-32	3403900	7
16	Screw 8-32 x 1/2	3427456	4
17	Hex nut 8-32	3405156	4
18	Gasket	8040700	1
19	Screw 3/8-16 x 1	3493900	1
20	Fender washer 3/8"	3478600	1
21	Base	8042921	1
22	Handle	8041800	1
23	Metal-sheath element	See Table 2	1

Table 2		
MODEL	VOLTS	METAL-SHEATH ELEMENT PART NUMBER
FHK-420-1A	208	04458502
FHK-424-1A	240	04458102
FHK-427-1A	277	04458602
FHK-448-1A	480	04461702
FHK-457-1A	600	04458902

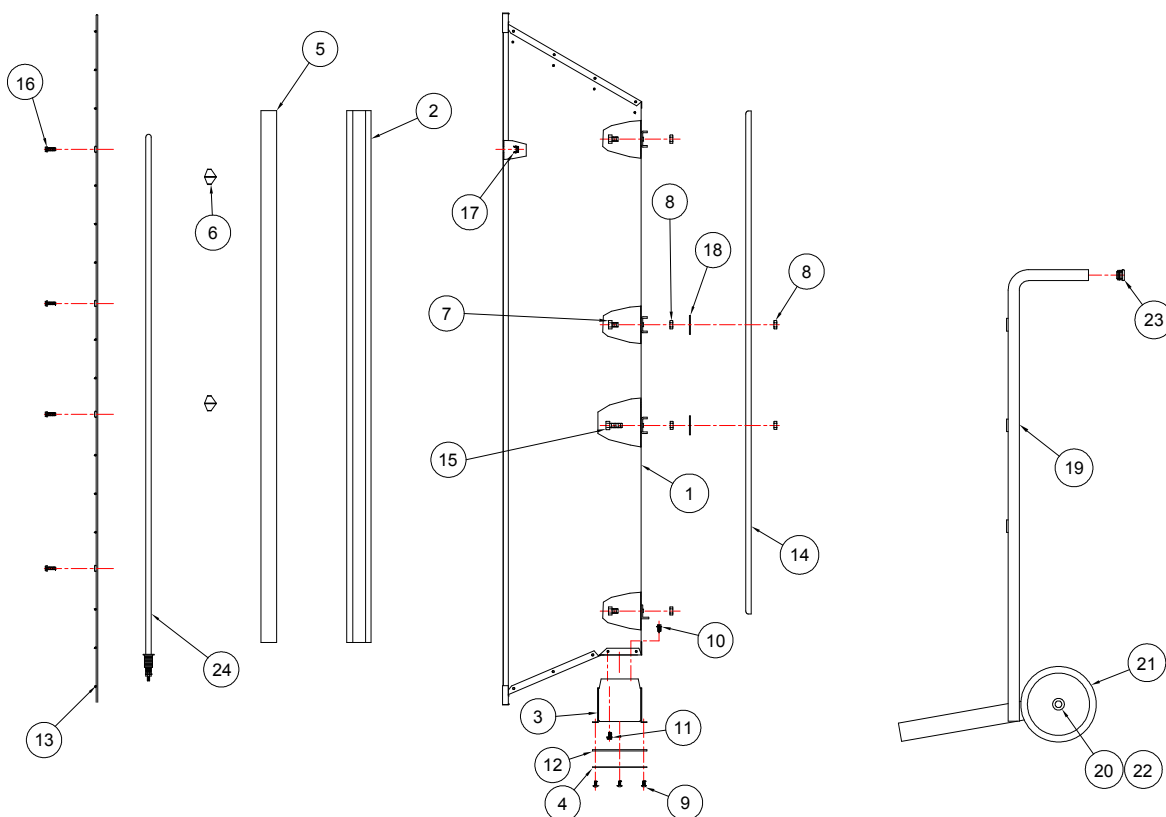
FHK-6 EXPLODED VIEW AND REPLACEMENT PARTS LIST



ITEM. NO.	DESCRIPTION	PART NUMBER	QTY.
1	Housing Assembly	8827999	1
2	Extrusion	8041700	3
3	Terminal box	8040802	1
4	Terminal box cover	8040900	1
5	Reflector	8048900	3
6	Metal sheath element support	8024100	3
7	Screw 3/8-16 x 1/2	3474300	2
8	Jam nut 3/8-16	3429100	10
9	Screw 8-18 x 1/2	3444456	4
10	Hex head screw 8-15 x 1/2	3470800	4
11	Hex head screw 10-32 x 3/8	3475000	1
12	Gasket	8041000	1
13	Lamp guard	8020829	1
14	Back shield	8020707	1
15	Screw 3/8-16 x 1	3493900	4
16	Screw 10-32 x 5/8	3424956	8
17	Self retaining nut 10-32	3403900	8
18	Fender washer 3/8"	3478600	4
19	Metal-sheath element	See Table 3	3
20	Cable 10-3 type S x 15' long *	3163601	1
21	Plug 3-pole 3-wire 30-50 amps *	3180000	1
22	3/4" cable connector *	7714124	1
23	Tubing plug	2415100	2
24	Cart weld assembly	3924402	1
25	Cart axle	8523632	1
26	Wheel	8043600	2
27	Push nut 1/2" diameter	8045700	2

Table 3		
MODEL	VOLTS	METAL-SHEATH ELEMENT PART NUMBER
FHK-620-3A	208	04458302
FHK-624-3A	240	04457902
FHK-624-1CA	240	04457902
FHK-627-3A	277	04458402
FHK-648-3A	480	04458002
FHK-657-3A	600	04458802

FHK-13 EXPLODED VIEW AND REPLACEMENT PARTS LIST



ITEM. NO.	DESCRIPTION	PART NUMBER	QTY.
1	Housing Assembly	8829599	1
2	Extrusion	8042400	3
3	Terminal box	8040802	1
4	Terminal box cover	8040900	1
5	Reflector	8049000	3
6	Metal sheath element support	8024100	6
7	Screw 3/8-16 x 1/2	3474300	8
8	Jam nut 3/8-16	3429100	16
9	Screw 8-18 x 1/2	3444456	4
10	Hex head screw 8-15 x 1/2	3470800	4
11	Hex head screw 10-32 x 3/8	3475000	1
12	Gasket	8041000	1
13	Lamp guard	8021929	1
14	Back shield	8021707	1
15	Screw 3/8-16 x 1	3493900	4
16	Screw 10-32 x 5/8	3424956	12
17	Self retaining nut 10-32	3403900	12
18	Fender washer 3/8"	3478600	4
19	Cart weld assembly	3924402	1
20	Cart axle	8523632	1
21	Wheel	8043600	2
22	Push nut 1/2" diameter	8045700	2
23	Tubing plug	2415100	2
24	Metal-sheath element	See Table 4	3

Table 4		
MODEL	VOLTS	METAL-SHEATH ELEMENT PART NUMBER
FHK-1320-3A	208	04458502
FHK-1324-3A	240	04458102
FHK-1327-3A	277	04458602
FHK-1348-3A	480	04461702
FHK-1357-3A	600	04458902

LIMITED WARRANTY

TPI Corporation – P.O. Box 4973 – Johnson City, TN 37602 – 1-800-251-0382

The warranty herein set forth is in lieu of all other warranties expressed or implied.

TPI Corporation of Johnson City, TN 37602, warrants its products to the owner against defects in material and workmanship for a twelve (12) month period under normal use and service following date of manufacture or installation when proof of such is provided to seller.

This Warranty requires that the owner, or his agent, install the equipment in accordance with the National Electrical Code, any other applicable heating or electrical codes and the manufacturer's installation instructions. It further requires that he perform reasonable and necessary maintenance on the unit. The company is not liable for abuse or misuse of product as may be finally determined by the company.

The obligation of TPI Corporation under the terms of this warranty, shall be to supply a new part, repair of defective part, or a refund, at the company's option with no cost to owner for the new or repaired part. Such parts are to be returned to the factory, or such other location as the company may designate. This warranty does not obligate TPI Corporation to bear the cost of labor in replacing any assembly, unit or component part thereof, nor does the company assume any liability for secondary charges, expenses for installing or removal, or any other consequential losses. The company's maximum liability shall not in any case exceed the list price for the product claimed to be defective.

The warranty shall not apply to any product or parts of products that have been repaired or altered outside of Seller's factory in any manner. Warranty is void on products that have been determined to have been subjected to misuse, negligence or accident or which have been used in a manner contrary to Seller's instructions.



MODEL & PART NO.'S:

FHK-622-1A (04859502) FHK-1322-1A (04859302)
FHK-638-3A (04859402) FHK-1338-3A (04859202)

WARNING! DO NOT ATTEMPT TO INSTALL, OPERATE, OR SERVICE THIS PRODUCT BEFORE READING ALL INSTRUCTIONS CAREFULLY. FAILURE TO COMPLY WITH THESE INSTRUCTIONS COULD RESULT IN FIRE, PERSONAL INJURY AND/OR PROPERTY DAMAGE! RETAIN INSTRUCTIONS FOR FUTURE REFERENCE.

DESCRIPTION

Fostoria portable heavy-duty radiant heaters are designed to provide electric infrared heat when permanent type heating is not practical or economical. They can be wheeled to the job and connected to disconnect switches or lighting centers. All units are clean in operation, and are weather resistant.

NOTE: The metal sheath heating element must be shielded from wind for maximum efficiency.

All models are supported on a steel handcart with two 6 inch semi-pneumatic wheels. These models have reflectors that can be adjusted to horizontal or vertical position for best heat pattern.

Protective wire guards are standard.

SPECIFICATIONS

6 KW UNITS

Power requirements:

FHK-622-1A.....220V (1PH: 27.3 Amps)

FHK-638-3A.....380V (3PH: 9.1 Amps)

Heat output.....6KW

Overall dimensions:.....37-3/4" H x 19-7/8" W x 16" D

Housing.....0.040 gold anodized aluminum

Reflector.....0.032 BOS aluminum

13 KW UNITS

Power requirement:

FHK-1322-1A.....220V (1PH: 61.4 Amps)

FHK-1338-3A.....380V (3PH: 20.5 Amps)

Heat output.....13.5KW

Overall dimensions:.....58" H x 27-1/2" W x 19" D

Housing.....0.040 gold anodized aluminum

Reflector.....0.032 BOS aluminum

REQUIRED CLEARANCE

WARNING: HAZARD OF FIRE: Minimum spacing from front of heater to combustible materials is:

4 feet for models: FHK-622-1A, 638-3A

7 feet for models: FHK-1322-1A, 1338-3A

IMPORTANT SAFETY INFORMATION

Read all instructions before using this heater.

1. Make certain that the power source conforms to the electrical requirements of the heater. Disconnect power before installing or servicing. If the power disconnect is out of sight, lock it in the open position and tag it to prevent unexpected application of power. Failure to do so could result in fatal electric shock.

WARNING: DO NOT DEPEND UPON A THERMOSTAT OR OTHER SWITCH AS THE SOLE MEANS OF DISCONNECTING POWER WHEN INSTALLING OR SERVICING THE HEATER. ALWAYS DISCONNECT POWER AT THE MAIN CIRCUIT BREAKER AS DESCRIBED ABOVE. FAILURE TO DO SO COULD RESULT IN FATAL ELECTRIC SHOCK.

2. This heater is intended ONLY for installation in accordance with the National Electric Code (NEC), all applicable local codes and ordinances, and all sections of this manual. Any variance voids the warranty and may create unsafe conditions. All wiring should be done by a qualified electrician, using copper wire only.
3. Special attention must be given to any grounding information pertaining to this heater. To prevent the risk of electrocution, the heater must be securely and adequately grounded. This should be accomplished by connecting a grounded conductor between the service panel and the green grounding screw provided on the heater. To ensure a proper ground, the grounding means must be tested by a qualified electrician
4. Do not insert fingers or foreign objects into the heater. Do not block or tamper with the heater in any manner while it is in operation. Do not touch heater while in operation or just after it has been turned off, as some parts may be hot enough to cause injury.

SAFETY INFORMATION (CONT.)

- Do not attach ductwork to this heater or attempt to use it as a make-up air heater. Such use voids the warranty and may create unsafe conditions.
 - This heater is intended for general heating applications ONLY. **IT MUST NOT BE USED IN POTENTIALLY DANGEROUS LOCATIONS SUCH AS FLAMMABLE, EXPLOSIVE, CHEMICAL-LADEN, OR WET ATMOSPHERES.**
 - In cases in which property damage may result from malfunction of the heater, a backup system or a temperature sensitive alarm should be used.
 - CAUTION: Check branch voltage. Heaters should be operated at voltages no higher than the heating elements and the heater are designed for. (Lower voltages, however, pose no problem other than a loss of wattage).
 - If metal sheath (metal rod element) has an obvious "bowing", an over-voltage/over-temperature condition exists. In case of a bowing condition, de-energize the heater immediately and contact the factory.
 - Heaters are not intended and are not safe for process heating applications.
 - This heater is not equipped with a tip-over switch or an over-temperature safety cutoff and therefore should not be used in close proximity to flammable materials.
 - Do not store in a damp location.
- WARNING: FAILURE TO FOLLOW ABOVE SAFETY INSTRUCTIONS COULD RESULT IN FATAL SHOCK HAZARDS!**

SAVE THESE INSTRUCTIONS!

WIRING INSTRUCTIONS

CAUTION: To protect against potential fire hazard, these heaters must be protected by one of the following fast acting fuses:

Gould-Shawmut
ATM A4J
A2K A6K

Buss
KTS-R KTN-R
KTK JKS

The fuse must be as close to the amps shown on the nameplate as possible.

- Remove the 4 machine screws holding the cover in place on the bottom of the terminal box.
- Select the proper cable size per Table A (page 3). Install a 3/4 inch cable connector (customer supplied).
- Use standard clamp type connectors, or if the heater is to be used outdoors or in damp locations, use liquid tight cable connectors. (Customer supplied)
- Connect cable to the 6-inch lead wires inside the terminal box, with wire nuts or other safe mechanical means.
- For proper cable size, see Table A on page 3
- Connect the heater to a suitable electrical circuit.

OPERATING INSTRUCTIONS

NOTE: It is normal for any high intensity heater to create some expansion or contraction noises during heat-up and cool-down cycles. These noises are nothing to be concerned about. Heaters are energized by placing the plug (customer supplied) into a grounded, matching receptacle, or wiring to an approved safety switch.

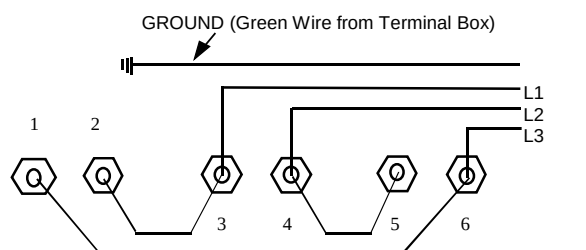


Figure 1
3-PHASE

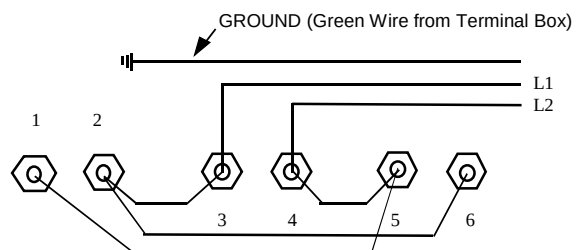


Figure 2
1 PHASE

ELEMENT REPLACEMENT

MAINTENANCE

WARNING: BE SURE TO TURN OFF POWER TO THE HEATER, OR REMOVE FUSES, TO ENSURE ANOTHER PERSON IN A REMOTE LOCATION CANNOT ENERGIZE THE HEATER WHILE REPLACING ELEMENTS OR PERFORMING ANY OTHER MAINTENANCE OPERATIONS.

1. Be sure heater is disconnected from power supply. Remove old element.
2. To replace an element, insert the U-shaped element through slots in the reflector and through holes in the terminal box.
3. Place gasket over element threads inside terminal box. Then replace metal washer and hex nut, and tighten.
4. Replace stakon to tab on the element, to connect the power supply. Replace rod support.
5. Use only Fostoria replacement elements of the same wattage/voltage and part number of the original element. (See Table A)

NOTE: A replacement inner reflector (reference no. 5) is included with every U-rod element replacement order. Install the new inner reflector when replacing elements.

Blow dust off the elements and inner reflectors prior to heating season, and more often in dusty environments. Repair or replace damaged or worn cord sets (cables) immediately.

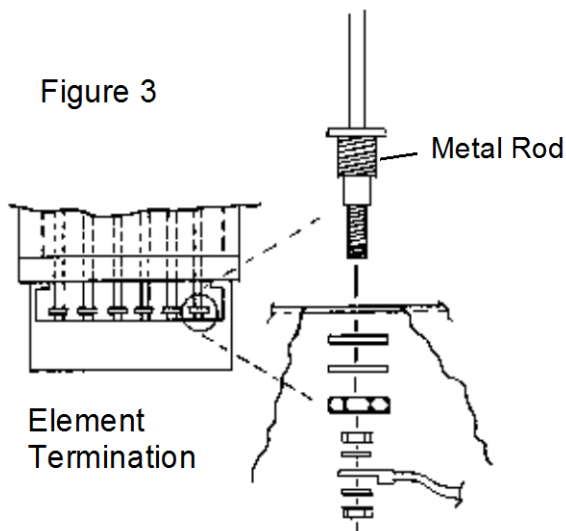
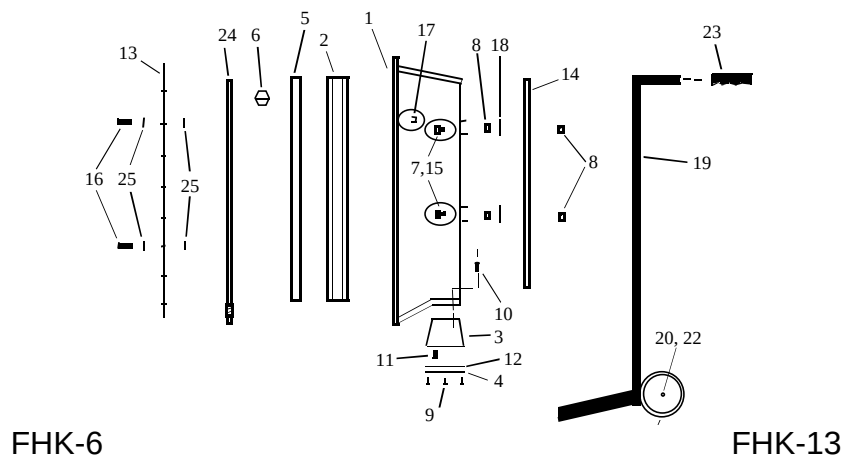


Table A

Heater Model	KW	VOLTS	PH	AMPS	Type S Cable Size*	Leadwire Item #26	Ground wire Item #27	Buss Bar Item #28	Element Item #24
FHK-622-1A	6	220	1	27.3	10/3	3160301	3160704	2415001	04860702
FHK-638-3A	6	380	3	9.1	12/4	3160633	3160704	2415000	04860802
FHK-1322-1A	13.5	220	1	61.4	6/3	6160301	3160704	2415001	04860902
FHK-1338-3A	13.5	380	3	20.5	12/4	6160633	3160704	2415000	04861002

* 50 ft. Maximum Cable Length

REPLACEMENT PARTS LIST and EXPLODED VIEW



REF. NO.	DESCRIPTION	PART NO.	QTY.	REF. NO.	DESCRIPTION	PART NO.	QTY.
1	Housing assembly	8827999	1	1	Housing assembly	8829599	1
2	Extrusion	8041700	3	2	Extrusion	8042400	3
3	Terminal box	8040802	1	3	Terminal box	8040802	1
4	Terminal box cover	8040900	1	4	Terminal box cover	8040900	1
5	Reflector	8048900	3	5	Reflector	8049000	3
6	Metal rod support	8024100	3	6	Metal rod support	8024100	6
7	3/8-16 x 1/2 hex hd screw	3474300	2	7	3/8-16 x 1/2 hex hd screw	3474300	8
8	3/8-16 jam nut	3429100	10	8	3/8-16 jam nut	3429100	16
9	8-32 x 1/2 hex hd washer screw	3444456	4	9	8-32 x 1/2 hex hd washer screw	3444456	4
10	8 x 1/2 hex hd screw	3470800	4	10	8 x 1/2 hex hd screw	3470800	4
11	10-32 x 3/8 hex hd screw	3475000	1	11	10-32 x 3/8 hex hd screw	3475000	1
12	Gasket	8041000	1	12	Gasket	8041000	1
13	Lampguard	8020829	1	13	Lampguard	8021929	1
14	Back shield	8020707	1	14	Back shield	8021707	1
15	3/8-16 x 1 hex hd screw	3493900	4	15	3/8-16 x 1 hex hd screw	3493900	4
16	10-32 x 5/8 hex hd screw	3424956	8	16	10-32 x 5/8 hex hd screw	3424956	12
17	10-32 self retaining nut	3403900	8	17	10-32 self retaining nut	3403900	12
18	3/8 fender washer	3478600	4	18	3/8 fender washer	3478600	4
19	Cart weld assembly	3924402	1	19	Cart weld assembly	3924402	1
20	Cart axle	8523632	1	20	Cart axle	8523632	1
21	Wheel	8043600	2	21	Wheel	8043600	2
22	Push nut 1/2 dia.	8045700	2	22	Push nut 1/2 dia.	8045700	2
23	Handle grip	8043700	2	23	Handle grip	8043700	2
24	U-shaped metal rod element	See Table A	3	24	U-shaped metal rod element	See Table A	3
25	10 flat washer	3400832	24	25	10 flat washer	3400832	24
26	Wire assy. BLK	See Table A	2	26	Wire assy. BLK	See Table A	2
27	Wire assy. GRN	See Table A	1	27	Wire assy. GRN	See Table A	1
28	Buss bar	See Table A	2	28	Buss bar	See Table A	2

WARRANTY

FOSTORIA INDUSTRIES, INC., 1200 N. Main St. - Fostoria, Ohio 44830 - (419) 435-9201

The warranty herein set forth is in lieu of all other warranties expressed or implied.

Fostoria Industries, Inc. of Fostoria, Ohio 44830, warrants its products to the owner against defects in material and workmanship for a twelve (12) month period under normal use and service following date of manufacture or installation when proof of such is provided to seller.

This Warranty requires that the owner, or his agent, install the equipment in accordance with the National Electrical Code, any other applicable heating or electrical codes and the manufacturer's installation instructions. It further requires that he perform reasonable and necessary maintenance on the unit. The company is not liable for abuse or misuse of product as may be finally determined by the company.

The obligation of Fostoria Industries, Inc. under the terms of this warranty, shall be to supply a new part, repair of defective part, or a refund, at the company's option with no cost to owner for the new or repaired part. Such parts are to be returned to the factory, or such other location as the company may designate. This warranty does not obligate Fostoria Industries, Inc. to bear the cost of labor in replacing any assembly, unit or component part thereof, nor does the company assume any liability for secondary charges, expenses for installing or removal, or any other consequential losses. The company's maximum liability shall not in any case exceed the list price for the product claimed to be defective.

The warranty shall not apply to any product or parts of products that have been repaired or altered outside of Seller's factory in any manner. Warranty is void on products that have been determined to have been subjected to misuse, negligence or accident or which have been used in a manner contrary to Seller's instructions.